Question

$\mathbf{X}$、$\mathbf{Y}$ are elements in the same group and have similar chemical properties. The dioxide of $\mathbf{X}$, $\mathbf{A}$, has widespread applications in organic chemistry, and its aqueous solution can be reduced by sulfur dioxide to generate the element $\mathbf{X}$ (reaction 1); $\mathbf{A}$ can react with $\mathbf{H}$ to obtain ion $\mathbf{B}$ and ion $\mathbf{C}$ (reaction 2), $\mathbf{B}$ has one more $\mathbf{X}$ atom than $\mathbf{C}$ and the charge remains unchanged. The chloride of $\mathbf{X}$, $\mathbf{D}$, reacts with NH_3 to produce an explosive substance $\mathbf{E}$ (reaction 3). $\mathbf{D}$ reacts with $\mathbf{A}$ to obtain $\mathbf{F}$, and $\mathbf{F}$ reacts with Cu element, and the products include $\mathbf{A}$ and $\mathbf{G}$ (reaction 4). $\mathbf{F}$ can also react with excess sulfur element (reaction 5), and the products include $\mathbf{A}$ and $\mathbf{G}$, as well as $\mathbf{I}$, which has a similar structure to $\mathbf{G}$. $\mathbf{Z}^{2-}$ containing 57.01% $\mathbf{Y}$ reacts with sulfite to generate the element $\mathbf{Y}$ and the oxygen-containing anion $\mathbf{H}$ (reaction 6). The sodium salt of $\mathbf{H}$ is a common reagent in analytical chemistry and reacts with I_2 to convert to $\mathbf{C}$.

Which of the following statements are correct?

1. $\mathbf{B}$ contains 11 atoms

2. Molecule $\mathbf{E}$ contains a 4-fold rotation axis

3. The ratio of reaction coefficients of each reactant and product in reaction 3 is 12:64:3:4:48

4. The ratio of reaction coefficients of each reactant and product in reaction 5 is 6:4:3:2:1

5. Sulfide ions are also produced in reaction 6

6. $\mathbf{I}$ is a reddish-brown gas

7. Reaction 2 does not require acid participation

8. Reaction 4 also produces a water-insoluble white precipitate.

A. All other options are incorrect

B. 2.3.6.

C. 1.2.3.4.6.

D. 1.4.5.

E. 2.3.5.

F. 3.6.8.

G. 2.4.7.8.

H. 1.5.8.

I. 2.4.7.

J. 2.3.

K. 4.5

L. 2.8.

Answer

Correct Answer: **D**

Detailed Explanation

According to the knowledge of elemental chemistry, $\mathbf{X}$ is Se, $\mathbf{Y}$ is Te

CHECKPOINT 1 PTS

$\mathbf{X}$ is Se, $\mathbf{Y}$ is Te

$\mathbf{A}$ is the dioxide of Se, so $\mathbf{A}$ is SeO_2; from the mass fraction, we can know that $\mathbf{Z}^{2-}$ should be TeS_3^{2-}. From the role of the sodium salt of $\mathbf{H}$ in analytical chemistry, we can know that $\mathbf{H}$ is $S_2O_3^{2-}$. Thus we can deduce that $\mathbf{C}$ is $S_4O_6^{2-}$, and from the fact that $\mathbf{B}$ has one more Se atom than $\mathbf{C}$, we can deduce that the chemical formula of $\mathbf{B}$ is $SeS_4O_6^{2-}$.

CHECKPOINT 1 PTS

$\mathbf{A}$ is SeO_2

CHECKPOINT 1 PTS

$\mathbf{Z}^{2-}$ is TeS_3^{2-}

CHECKPOINT 1 PTS

$\mathbf{H}$ is $S_2O_3^{2-}$

CHECKPOINT 1 PTS

$\mathbf{C}$ is $S_4O_6^{2-}$

CHECKPOINT 1 PTS

$\mathbf{B}$ is $SeS_4O_6^{2-}$

The chloride of Se, $\mathbf{D}$, should be $SeCl_4$, which corresponds to $\mathbf{E}$ when reacted with ammonia to yield Se_4N_4.

CHECKPOINT 1 PTS

$\mathbf{D}$ should be $SeCl_4$

CHECKPOINT 1 PTS

$\mathbf{E}$ is Se_4N_4

$\mathbf{D}$ reacts with $\mathbf{A}$ to yield $\mathbf{F}$, which should be $SeOCl_2$, and can react with Cu to yield $\mathbf{G}$ as Se_2Cl_2, and reacts with sulfur to yield $\mathbf{I}$ with a similar structure: S_2Cl_2.

CHECKPOINT 1 PTS

$\mathbf{F}$ is $SeOCl_2$

CHECKPOINT 1 PTS

$\mathbf{G}$ is Se_2Cl_2

CHECKPOINT 1 PTS

$\mathbf{I}$ is S_2Cl_2

$\mathbf{B}$ contains 4+6+1=11 atoms, 1 is correct.

The molecule $\mathbf{E}$ contains a 4-fold rotoinversion axis but not a rotation axis, 2 is incorrect.

$\mathbf{I}$ is a red-brown liquid, 6 is incorrect.

Reaction 1. $SeO_2 + 2SO_2 + 2H_2O = 2H_2SO_4 + Se$

CHECKPOINT 1 PTS

Reaction 1. $SeO_2 + 2SO_2 + 2H_2O = 2H_2SO_4 + Se$

Reaction 2. $4H^+ + SeO_2 + 4S_2O_3^{2-} = S_4O_6^{2-} + SeS_4O_6^{2-} + 2H_2O$

Reaction 2 requires acid participation, 7 is incorrect.

CHECKPOINT 1 PTS

Reaction 2. $4H^+ + SeO_2 + 4S_2O_3^{2-} = S_4O_6^{2-} + SeS_4O_6^{2-} + 2H_2O$

Reaction 3. $12SeCl_4 + 64NH_3 = 3Se_4N_4 + 2N_2 + 48NH_4Cl$

The ratio of the reaction coefficients of each reactant and product in reaction 3 is 12:64:3:2:48, 3 is incorrect.

CHECKPOINT 1 PTS

Reaction 3. $12SeCl_4 + 64NH_3 = 3Se_4N_4 + 2N_2 + 48NH_4Cl$

Reaction 4. $3Cu + 4SeOCl_2 = 3CuCl_2 + 2SeO_2 + Se_2Cl_2$

$CuCl_2$ is soluble in water, 8 is incorrect.

CHECKPOINT 1 PTS

Reaction 4. $3Cu + 4SeOCl_2 = 3CuCl_2 + 2SeO_2 + Se_2Cl_2$

Reaction 5. $6S + 4SeOCl_2 = 3S_2Cl_2 + 2SeO_2 + Se_2Cl_2$

The ratio of the reaction coefficients of each reactant and product in reaction 5 is 6:4:3:2:1, 4 is correct.

CHECKPOINT 1 PTS

Reaction 5. $6S + 4SeOCl_2 = 3S_2Cl_2 + 2SeO_2 + Se_2Cl_2$

Reaction 6. $TeS_3^{2-} + 2SO_3^{2-} = 2S_2O_3^{2-} + S^{2-} + Te$

CHECKPOINT 1 PTS

Reaction 6. $TeS_3^{2-} + 2SO_3^{2-} = 2S_2O_3^{2-} + S^{2-} + Te$

Sulfide ions are also generated in reaction 6, 5 is correct.

It can be found that only 1, 4, and 5 are correct, choose D.